

Speak Like A Mathematician:

included angle:

included side:

hypotenuse:

leg:

congruence criterion:

The Five Triangle Congruence Criteria

SSS: if three _____ of one triangle are congruent to three sides of another, the triangles are congruent.

SAS: two sides and the _____ angle. **ASA:** two angles and the _____ side.

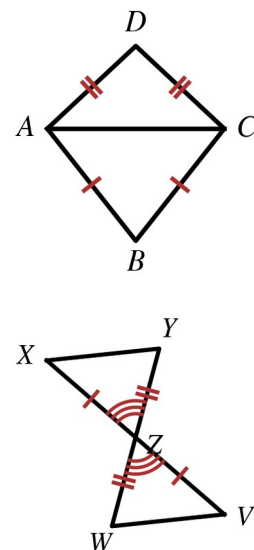
AAS: two angles and a _____ side. **HL:** in RIGHT triangles, the _____ and one _____.

There is no SSA and no AAA. Same angles can mean a bigger copy; SSA can leave _____ different triangles.

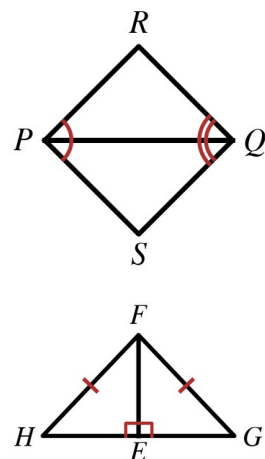
Example 1 - Name the Criterion

Each pair of triangles at the right is marked. Name the criterion that proves them congruent.

- a) $\triangle ABD$ and $\triangle CBD$
- b) $\triangle XYZ$ and $\triangle VWZ$
- c) $\triangle PQR$ and $\triangle PQS$
- d) $\triangle EFG$ and $\triangle EFH$



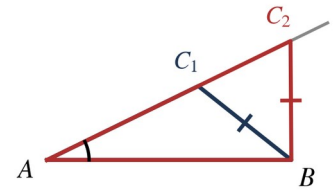
SC Honors Geometry - Nappier
Triangle Rigidity — SSS, SAS, ASA, AAS, HL



Example 2 - Why SSA and AAA Are Not Criteria

Study the two counterexamples at the right.

- In the first figure, which three parts are congruent in both triangles? Why is that not enough?
- In the second figure, all three angles are congruent. Why are the triangles not congruent?
- Which criteria DO use two sides and an angle? How are they different from SSA?

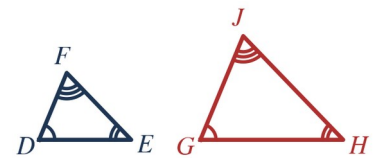


same two sides and the same non-included angle —
but two different triangles

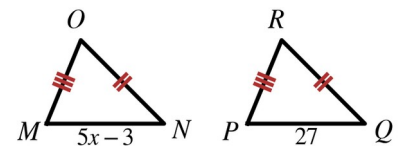
Example 3 - Find the Value of x

In the diagram, $\triangle MNO \cong \triangle PQR$.

- Which side corresponds to $\overline{MN}$?
- Write and solve an equation for x.
- Check your answer.



all three angles congruent — but not the same size

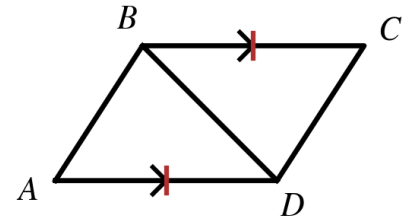


Example 4 - Proof Using SAS

Given: $\overline{BC} \parallel \overline{AD}$, $\overline{BC} \cong \overline{AD}$

Prove: $\triangle ABD \cong \triangle CDB$

Statements	Reasons
1. $\overline{BC} \parallel \overline{AD}$	1. Given
2.	2.
3.	3.
4.	4.
5.	5.

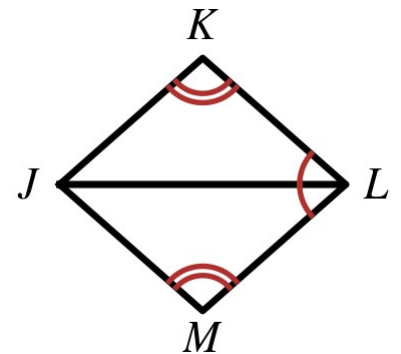


Example 5 - Proof Using AAS

Given: $\overline{JL}$ bisects $\angle KLM$; $\angle K \cong \angle M$

Prove: $\triangle JKL \cong \triangle JML$

Statements	Reasons
1. $\overline{JL}$ bisects $\angle KLM$	1. Given
2.	2.
3.	3.
4.	4.
5.	5.

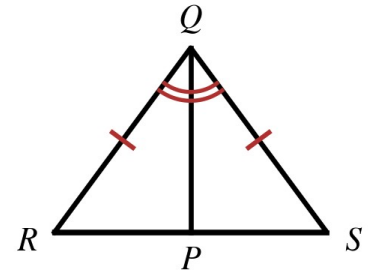


Example 6 - Proof Using SAS Again

Given: $\overline{QP}$ bisects $\angle RQS$; $\overline{QR} \cong \overline{QS}$

Prove: $\triangle RQP \cong \triangle SQP$

Statements	Reasons
1. $\overline{QP}$ bisects $\angle RQS$	1. Given
2.	2.
3.	3.
4.	4.
5.	5.



Example 7 - Choose a Criterion

Decide whether the given information is enough to prove the triangles congruent. If it is, name the criterion.

- a) Two pairs of sides and the pair of angles between them.
- b) Two pairs of angles and the pair of sides between them.
- c) Two pairs of angles and a pair of sides NOT between them.
- d) Two pairs of sides and a pair of angles NOT between them.
- e) All three pairs of angles.